Lightweight Multiscale Feature Fusion and Multireceptive Field Feature Enhancement for Small Object Detection in the Aerial Images

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***Abstract*—Small uncrewed autonmous vehicles (UAVs) equipped with deep learning models are increasingly used to detect small objects both on the ground and in aerial environments. Since small objects occupy fewer pixels in the images, deep learning models capable of effectively extracting their features demand significant computational resources. However, the limited computational capacity of lightweight UAVs results in lower object recognition accuracy. Existing lightweight methods reduce the parameters by using lightweight convolution or an optimized feature fusion network architecture, but the former sacrifices the ability to capture the detailed features of small objects, and the latter shows a significant negative correlation between the number of parameters and detection accuracy. Aiming at the problem that it is difficult to balance the model parameters and the detection accuracy in a complex working environment, this article proposes a lightweight small object detection algorithm [lightweight multiscale feature fusion and multireceptive field feature enhancement (LMFF-MFFE)] based on multiscale feature fusion and multireceptive field feature enhancement. First, this article introduces a lightweight multiscale dense feature fusion network (MDFFN), which reduces the parameters while enriching the object feature representation by trimming and optimizing the multiscale dense feature propagation path. Second, a multireceptive field feature enhancement (MRFFE) module is integrated to capture local contextual information around objects to further enhance the effectiveness of fused multiscale features. Experimental results demonstrate superior performance on benchmark datasets: LMFF-MFFE achieves 44.5% mAP (the average precision value) on VisDrone2019 and 92.6% mAP on NWPU VHR-10, and the number of parameters is reduced by 20.4%, outperforming both baseline models and mainstream methods. The proposed LMFF-MFFE algorithm effectively balances computational efficiency and detection accuracy under resource-constrained UAV platforms, showing particular advantages in complex environments.**

***Index Terms*—Dilated convolution, feature fusion, lightweight, small object detection.**

I. Introduction

AUTONOMOUS environmental perception represents a critical research direction in remote sensing, where object detection serves as an important prerequisite for scene understanding and visual analysis by effectively acquiring external information [1]. Currently, object detection methods are primarily categorized into two types: two-stage detectors and single-stage detectors. The two-stage approach initially generates region proposals containing potential objects, followed by classification and precise localization. However, this region proposal generation process incurs significant computational overhead during both training and inference phases. Representative two-stage detectors include RCNN [2], Fast RCNN [3], Faster RCNN [4], and Mask RCNN [5]. In contrast, single-stage detectors perform direct feature extraction from input images, simultaneously predicting bounding box coordinates and class probabilities. This architecture achieves faster inference speeds and superior computational efficiency compared to two-stage methods, making it particularly advantageous for real-time detection scenarios. Notable single-stage implementations encompass the YOLO series [6], [7], [8] and the single-shot MultiBox detector (SSD) [9]. Among them, YOLOv8 series algorithms show excellent detection performance. YOLOv8s [10], [11] has a rich feature expression capability, and its network architecture retains the efficient detection of previous YOLO series while further optimizing the backbone network and the design of the detection head, making it possible to accurately and quickly locate and identify objects in complex scenes.

Due to the dense distribution and serious mixing of small objects in input images, existing two-phase and one-phase object detection methods repeatedly downsample images through convolution operations during feature extraction. This progressive reduction in feature map resolution hinders the model's ability to extract fine-grained details, resulting in inaccurate detection of small objects. Consequently, these methods struggle to adapt to multiscale detection tasks. To address this, researchers proposed feature pyramid networks (FPNs) [12], which leverage a hierarchical structure: higher level feature

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maps detect larger objects, while lower level feature maps focus on small objects. By integrating a multiscale pyramid with a top–down fusion strategy, FPN retains both high-level semantics and fine-grained details, significantly enhancing multiscale detection capabilities.

Since then, researchers have explored two main directions to enhance FPN. The first approach focuses on extending the pyramid architecture to enrich feature representations. By stacking convolutional blocks and integrating attention mechanisms, these techniques comprehensively fuse multiscale features, thereby improving the representational capacity of the network. Wang et al. [13] mitigated the problem of information degradation that often occurs during the transmission of small object features by building a cross-scale feature interaction pyramid network. Deng et al. [14] proposed an extended pyramid network with an embedded fast texture transfer (FTT) module to capture richer spatial details for small object detection. Zang et al. [15] introduced a dense convolutional interaction module that iteratively fuses multilevel features, effectively mitigating information loss across pyramid levels. However, such architectural expansions significantly increase model parameters and computational complexity, raising the risk of overfitting on small datasets and hindering deployment on resource-constrained platforms like uncrewed autonmous vehicles (UAVs). In practical applications, lightweight models are crucial for resource-constrained devices such as UAVs, as they reduce computational and storage demands while improving deployment efficiency without compromising utility. The second approach aims to enhance feature discriminability by deeply mining feature information and complementarily fusing feature details, thereby reducing missed or false detections. Xie et al. [16] introduced spatial and channel complementary modules to enhance the useful information and suppress irrelevant information in the process of feature fusion to improve the robustness of the model. Ding et al. [17] proposed an adaptive weighted feature fusion method based on selective pooling, which assigns fusion weights by estimating the ratio of the number of objects between neighboring layers. However, under limited training data, such models tend to overfit to noisy patterns, ultimately degrading their generalization capability.

While the aforementioned methods enhance object detection performance to some extent, their excessively dense connectivity results in overly complex architectures and a substantial increase in model parameters, thereby hindering deployment on resource-constrained platforms like UAVs. For example, in dense fusion networks, each feature map layer is interconnected with multiscale counterparts. As network depth grows, these layers receive inputs from an expanding number of scales, exponentially increasing computational demands during fusion. To improve fusion effectiveness, integrated attention mechanisms often employ fully connected layers. The computation of attention weights introduces additional parameters, further escalating computational complexity. Although the effectiveness of the above methods is improved, the training cost increases accordingly, making network optimization difficult.

To address the above challenges, this article proposes a lightweight multiscale feature fusion and multireceptive field feature enhancement (LMFF-MFFE), which enhances detection accuracy while reducing model parameters, thereby achieving an optimal balance between efficiency and performance. In LMFF-MFFE, a lightweight multiscale dense feature fusion network (MDFFN) architecture is designed to efficiently fuse shallow fine-grained details (e.g., object contours and edges) with deep semantic information (e.g., object behavioral states) while reducing the number of model parameters. By incorporating multiple skip connections and cross-layer dense connections, the proposed framework fully integrates feature maps at each level. This approach not only prevents information loss but also enhances the representational power of the network by interactively merging shallow detail features and deep semantic features, thereby generating more informative feature representations for downstream detection tasks and the number of model parameters is reduced by cropping out the less important feature map propagation paths without degrading the object detection performance. To further enhance the multiscale applicability of the fused features and improve the detection capability, this article upgrades the RFB module into the multireceptive field feature enhancement (MRFFE) module, streamlining its branch structure to lower the computational cost and enlarging dilation rate of dilatation convolution to obtain richer contextual information of the object. The main contributions of this article are summarized as follows.

1) *Integrated lightweight detection framework (LMFF-MFFE):* This article introduces a single-stage architecture for small-object detection on resource-constrained airborne platforms. The framework jointly handles feature extraction, fusion, and context enhancement without relying on separate postprocessing stages.
2) *Multiscale dense feature fusion network:* MDFFN removes redundant cross-scale connections while retaining bidirectional channel links, preserving rich multiscale information flow with a reduced parameter and computation footprint.
3) *Dynamic multireceptive-field feature enhancement module:* MRFFE adopts a lightweight multibranch dilated-convolution strategy that adaptively adjusts receptive fields, strengthening local context and improving discrimination of densely distributed small objects without adding extra parameters.
4) *Balanced accuracy-efficiency trade-off across diverse aerial datasets:* The combined LMFF-MFFE design consistently improves performance over recent lightweight detectors while operating within the limited memory and compute budgets typical of onboard UAV hardware, indicating good generalization and deployability.

II. Related Work

In object detection imagery, small objects typically exhibit small size, low resolution, and recognition, causing their features to be inconspicuous under complex shooting conditions. Therefore, improving the detection of small objects has become an urgent challenge in computer vision. A widely

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adopted solution to improve the detection effect of small objects is multiscale feature enhancement. Building on the concept of feature enhancement, researchers have proposed some effective improvement algorithms from the two directions of feature pyramid fusion network and multiscale feature extraction network.

A. Pyramid Fusion Networks for Feature Enhancement

In CNNs, shallow local feature maps typically have higher spatial resolution and contain detailed information such as object edges, texture, and color. In contrast, deep global features have lower spatial resolution but provide rich semantic information, including object shapes, structures, and categories. Therefore, to enhance the feature representation of the object, small object detection networks usually fuse shallow and deep features to achieve improved detection performance. There are two strategies for early multiscale fusion: one is to obtain feature maps with different receptive field sizes through convolutional kernels of different sizes [18], and the other is to detect objects of different sizes by inputting images of different scales through an image pyramid [19]. However, both approaches typically demand extensive computational resources and high-performance hardware, limiting their practical utility and applicability [20].

Lin et al. [12] first proposed the FPN, which fuses shallow high-resolution features and deep semantic information through top–down paths and horizontal connections to achieve across-layer information propagation, and uses upsampling and element-wise summation to generate feature representations with rich multiscale information. However, FPN only uses top–down path propagation of features, resulting in deep feature maps that lack detailed information about the object, which is suboptimal for multilevel feature fusion. To enhance the FPN, Liu et al. [21] proposed PAFPN by adding a bottom–up propagation path to complement FPN, which passes the low-level feature information to the high-level feature maps and allows the low-level precise localization information to pass upward efficiently, to better integrate the multi-scale feature maps and further enhance the localization representation of the feature pyramid. However, when each feature layer is added to the information propagation path, additional convolutional layers need to be introduced and parameterized to further process and integrate the information, which directly leads to a significant increase in the model parameters, which brings a greater computational burden and memory requirements for model training and deployment, and the method lacks the distinction of the importance of the output feature maps of different feature layers in the process of feature fusion. Specifically, the output feature maps of each layer are weighted with the same contribution weights, without considering the differences in the different feature layers in capturing the object information and the differences in the importance, which prevents the feature fusion from being optimal and may affect the final object detection performance. Based on the limitations of this method, Tan et al. [22] proposed a BiFPN, which optimized the network structure of PAFPN so that each layer in the feature pyramid receives information from all other layers and dynamically adjusts the importance of different input features in a weighted manner, enhancing the feature representation. Ghiasi et al. [23] proposed NAS-FPN, which also built upon the optimized PAFPN by incorporating neural network architecture search (NAS) to find the optimal cross-scale connections. However, the search cost NAS requires to obtain the optimal object architecture is high, and the architecture offers limited extensibility.

Existing pyramid networks can fuse multiscale feature information through bottom–up and top–down paths. The YOLOv8s algorithm also introduces a bidirectional FPN to bring advantages for small object detection, based on which researchers optimize to preserve shallow feature map details and improve the accuracy of small object detection [24], [25]. However, since small objects are often surrounded by complex background noise, their features can easily be ignored or confused during shallow local and deep global feature fusion. This issue underscores the need for novel methods to enhance the saliency of small object information. Zhu et al. [26] proposed the GMDRA-net small object detection method, which increases the local details of global perception by combining the self-attention mechanism with linear transformation convolutional mapping to obtain finer global information and adopting a sparse strategy to dynamically perceive the intrinsic connection of features, eliminating the influence of noise and the phenomenon of omission of detection that exists in the dense distribution of small objects. Li et al. [27] introduced a global and local attention mechanism based on the PAFPN network architecture. This mechanism combines local feature analysis with global contextual information processing to enable the model to capture both complex object details and global structure in an image, thus improving the model's ability to interpret complex images. Wang et al. [28] proposed the SCFNet network to add additional local and global features to the pyramid structure. Feature aggregation is performed for every two adjacent layers during the bottom–up path transmission of the pyramid. These additional feature propagation paths are then aggregated with the final feature output layer of the same scale, thus increasing the richness of small object information. Liu et al. [29] introduced the CATNet network, which aims to aggregate visual features from all the different feature extraction backbone network stages, with cross-layer dense connectivity for both top–down and bottom–up paths, allowing the network to generate more consistent and robust feature representations. Regarding feature propagation path design, both SCFNet and CATNet increase the feature propagation paths without changing the original pyramid architecture, thereby improving the detection of multiscale objects. Jiang et al. [30] proposed BDFPN, which horizontally extends the scale of the FPN to enhance the fusion of feature information and introduces multiple skip connections to ensure feature fusion while the effective transfer of feature information is carried out, realizing the effective fusion of shallow and deep feature information. In addition, inspired by the successful application of U-net networks and nested structures in medical image segmentation, the researchers designed the dense nested interaction module (DNIM) [31]. This module allows the model to learn finer grained feature representations by setting multiple nodes on the path between the encoder and decoder sub-

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networks, which are connected to form a nested heteromorphic network to achieve progressive interaction between global and local features.

In terms of improving feature richness, the above methods rely on highly dense connectivity, which increases network complexity and inflates the number of parameters, thereby complicating model stability and optimization. To reduce the network complexity and achieve real-time small object detection on UAV platforms, Li et al. [32] optimized the feature fusion network architecture and designed a medium-density feature fusion module containing only three feature layers as the network backbone, which connects and fuses all feature information and promotes the reuse of features. Lu and Sun [33] replaced the ordinary convolution in the backbone network with a lightweight convolutional SPDConv, which reduces the computational complexity while preserving the fine features of small objects. Similarly, Sun et al. [34] applied lightweight convolutional GhostConv to reconstruct the network to reduce model complexity. Yang et al. [35] combined PConv with neural network architecture search to allow the network to select the most appropriate structure at different levels based on demand, thus improving the reduction of redundant computation while maintaining performance. Qi et al. [36] proposed a SODNet detection network, which reduces the time complexity in the feature extraction process by performing parallel feature extraction on multichannel images and reduces the processing time of multiscale feature map fusion by using nearest-neighbor interpolation and sub-pixel convolution to achieve fast upsampling. Although the above methods have achieved some results in reducing network complexity and realizing lightweight detection of small objects, there are still some problems. Some methods sacrifice small-object feature information while reducing complexity, leading to a decrease in detection accuracy, and the use of lightweight convolution reduces the computational effort but has limited ability to capture small-object features in complex scenes. To address the above problems, we focus our perspective on the feature fusion network architecture and propose an improved FPN that better balances the relationship between the number of parameters and the detection accuracy by reducing the model parameters by cropping and optimizing the dense feature map propagation links, and improving the detection performance by adding an appropriate amount of cross-layer features and bidirectional propagation paths.

B. Multibranch Extraction Network for Feature Enhancement

The pyramid model accommodates feature extraction and object detection for both large and small objects by constructing multiscale feature maps. However, in complex scenarios involving irregularly shaped objects or occluded objects, the limited receptive field often hinders the capture of comprehensive contextual information, resulting in insufficient data for accurate classification and localization [37]. Therefore, increasing the receptive field allows the model to focus on a larger image area around the object, increase the understanding of the relationship between the object and the surrounding environment, and improve the detection accuracy of various types of objects [38], [39]. In data processing, the pyramid model [40] is mainly responsible for multiscale feature construction and preliminary fusion, providing different levels of feature representation for subsequent processing, while the receptive field enhancement module [41] is based on the features output from the pyramid model, which further enhances the expression ability of the features and makes up for the deficiencies in the receptive field of the pyramid model, and the two collaborate to improve the detection performance of the whole small object detection system for different scales and complex objects. The two work together to improve the detection performance of the whole small object detection system for different scales and complex objects.

Conventional CNNs usually use a fixed-size convolutional kernel for feature extraction and downsampling the feature map to obtain multiscale feature information of the object [42]. However, there are two technical loopholes in this approach. First, it is difficult to adapt to multiscale object detection due to the fixed size of the convolutional kernel. Second, for small-size objects, the feature information will be gradually lost with multiple downsampling of the feature map, making it challenging to capture tiny object features. Multibranch flow convolution as an effective strategy can improve detection performance by extracting multiscale and multilevel features through different sizes of convolution kernel branches [43]. Therefore, Jiang et al. [30] employed convolution operations with convolution kernel sizes of 1×1, 3×3, and 5×5 in a multibranching structure to extract feature information with multiple receptive field sizes, which captures the relationship between pixels in the feature map and improves the detection accuracy. Zhang et al. [44] proposed a multiscale feature extraction module built upon the residual module, which has 5×5 and 3×3 convolutional filter branches to enhance the network's multiscale perception capability. Cai et al. [45] extracted dense texture features by arranging four different-sized depth convolutional kernels of 5×5, 7×7, 9×9, and 11×11 in parallel and adaptively fused the different features along the channel dimensions to enable the collection of local contextual information. The above methods enhance the representation of feature information by employing different-sized convolutional kernels to obtain the contextual information of the object, enabling them to improve detection accuracy.

However, the large number of parameters brought by ordinary convolution stacking makes it challenging to apply to complex airspace scenarios with limited resources. Therefore, the introduction of dilation convolution by inserting a certain number of zero-valued elements between the elements of the standard convolution kernel, referred to as dilation, expands the receptive field without adding additional parameters, which is more helpful for the model to capture the information of dense small objects and their surrounding contexts, and improves the accuracy of detection. Liu et al. [46] adopted this idea by constructing an RFB module to simulate the receptive field of human vision, combining it with multibranch expansion convolution to enhance the feature extraction capability of the network. Shang et al. [47] replaced the 3×3 convolutional layers with 1×3 and 3×1 convolutional layers on the top of RFB, which reduces the computation by one-third compared to

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the original network. Inspired by the idea of RFB, Li et al. [48] designed convolutional sub-branches with different expansion rates to capture different receptive field feature maps, and then spliced them with the original feature maps to obtain new feature maps with deep semantics and multiple receptive fields, which are sufficient to take into account both the local details and the global information to improve the detection effect. Min et al. [49], dynamically control the receptive fields of small and medium-sized objects based on multibranch expansion convolution. They employed the attention score at the lower layers to determine the importance of channels with multiple receptive fields. This dynamic adjustment mechanism helps to assist the network to improve the detection accuracy of small and medium-sized objects in the image. However, it requires an additional amount of computation to compute the attention score. Liu et al. [50] found that the contextual information corresponding to different receptive fields cannot be consistently well applied in all different scenes involving tiny objects. Based on this problem, they proposed a multifield adaptive feature enhancement module; by fusing the features of different receptive fields, the model can obtain a more comprehensive feature representation and improve the comprehension of complex scenes, but the fixed stream branching structure cannot be adapted to all complex detection scenarios, and a larger convolution kernel still introduces more computational parameters. In summary, many researchers inspired by different ideas have proposed various types of methods based on dilated convolution, which have positive effects in enhancing feature extraction capability, capturing multiscale information, but generally have the disadvantage of computational complexity, and it is difficult to find a balance between computational complexity and detection accuracy.

Therefore, to address the gaps above, this article introduces a lightweight MRFFE module, which contains a multibranch structure of standard convolution and expansion convolution. Considering the importance of the number of parameters for network optimization, this module enables the network to obtain richer local object context information without increasing the number of parameters and improves detection performance.

In summary, to address the problem that the overly dense connection of the multiscale feature pyramid structure leads to complex network structure, proliferation of the number of model parameters, difficulty in meeting the stability, and increased optimization difficulty, this article proposes an improved pyramid structure, which reduces the model parameters by optimizing the feature map propagation link, and achieves an effective balance between the number of parameters and the detection quality while improving the detection performance. Second, by integrating the MRFFE module, the FPN is assisted to enhance the multiscale applicability of the fused features, which further improves the object detection performance.

III. Methods

A. Architecture

As shown in Fig. 1, the proposed model LMFF-MFFE takes the aerial optical image as input, sequentially performs feature extraction in the backbone network, multiscale dense feature fusion, and feature enhancement, and finally generates the detection results.

1) Backbone Feature Extraction: The input image is preprocessed before fed into the backbone network. YOLOv8 is adopted as the backbone because it supports lightweight deployment while maintaining high accuracy. The YOLOv8n variant, although more compact, lacks sufficient feature extraction capacity and performs poorly on small objects. By contrast, YOLOv8s can significantly improve small-object feature retention and overall detection precision, yet increases the parameter count only marginally. Consequently, YOLOv8s provides the best trade-off between model size and accuracy for our task. To achieve an effective balance between the number of model parameters and the detection accuracy, in this article, the backbone network of YOLOv8s [51] is selected to perform the feature extraction process, and the extracted multiscale feature maps are fed into the feature fusion network. The module is shown as backbone in Fig. 1, where C_i denotes feature layers of different scale sizes obtained after convolution operation.

2) Multiscale Dense Feature Fusion: To compensate for the feature information loss problem caused by multiple downsampling, a bidirectional feature pyramid fusion network is usually used to fuse the deep semantic information with the shallow detail information, or multiple feature propagation paths are introduced to repeat the fusion based on this network, but this is accompanied by a sharp increase in the model complexity and a dramatic increase in the number of parameters. Therefore, this article proposes a lightweight MDFFN architecture, which fully integrates feature information of objects at different scales based on reducing the number of parameters. Multilevel features are repeatedly fused with feature maps of different scales through cross-layer and dense connections. In the fusion process, learnable weights are introduced so that the model is adaptively adjusted according to the importance of different features to ensure that the features are effectively utilized. The module is shown as a MDFFN in Fig. 1, where C_i' denotes the multiscale feature map generated after the first input fusion in the top–down process, and C_i'' denotes the multiscale feature map generated after the second input fusion in the bottom–down process. Detailed information is presented in Section III-B.

3) Feature Enhancement: To assist the network in improving the detection accuracy of objects with different densities, contextual information is needed to enhance the differentiation between objects. This article introduces an improved MRFFE module before object detection, enabling the model to include richer spatial information related to dense small objects by dynamically adjusting features from different receptive fields. This module is shown as MRFFE in Fig. 1, and detailed information will be presented in Section III-C.

4) Object Detection: The enhanced feature map is fed into the detection end, and the object detection results, such as feature categories, confidence, and bounding box information, are obtained by convolution. Based on the architecture of YOLO [52], the predictive output head usually uses P_6, P_5, and P_4 with a resolution of 20×20, 40×40, and 80×80 as the

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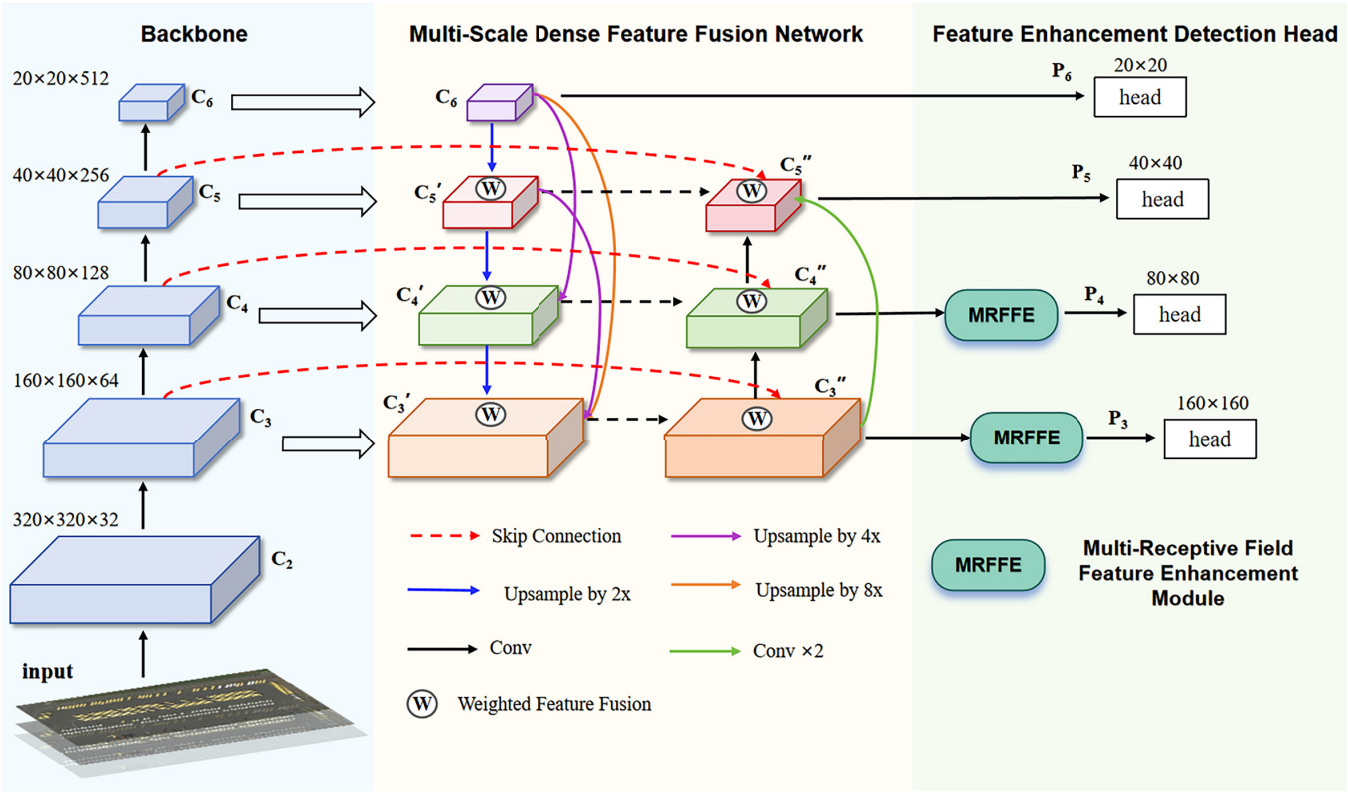


Fig. 1. Overall LMFF-MFFE architecture.

main output of the final detection result. However, it is difficult to comprehensively detect dense small objects under complex operating environments. Therefore, to improve the detection effect of small objects, this article, based on the retention of the basic detection head, adds a prediction output head P_3 with a resolution of 160×160 at the detection end, which improves the classification and localization accuracy of small objects and ensures comprehensive object detection. The module is shown as feature enhancement detection head in Fig. 1.

B. Multiscale Dense Feature Fusion Network

Multiscale feature map information dissemination aims to fuse feature information from different stages of the backbone network and disseminate it among different layers to improve the richness of feature information in the feature pyramid.

The overall architecture of the MDFFN network described in this article is shown in the yellow background of Fig. 1, where rectangles of different sizes represent the feature maps of a downsampling stage and the lines represent the feature information transmission streams. The MDFFN takes the network input feature pyramid ($C_2 \sim C_5$) as the input, aggregates it through top–down and bottom–up bidirectional feature propagation paths, and employs the interlayer residual connection (red dashed line in Fig. 1 MDFFN), cross-layer dense connections (green, purple, and orange solid lines in Fig. 1 MDFFN), and feature reweighting strategy (Fig. 1 W symbol in MDFFN) to fuse the feature maps.

The detailed feature map fusion process is shown in Fig. 2. In the top–down path, parameter transformation is performed

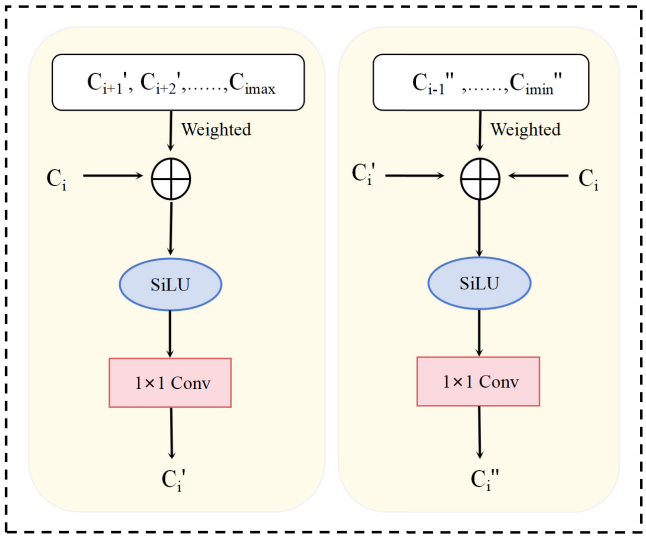


Fig. 2. Detailed fusion process of MDFFN feature map.

after fusing the information of the current layer and all the upper layers of the feature maps to obtain the output features under the path, specifically, layer C_5 generates the fused feature map C_5' by fusing the upsampled layer C_6, C_4 generates the fused feature map C_4' by fusing the upsampled layers C_6 and C_5', C_3 generates the fused feature map C_3' by fusing the upsampled layers C_6, C_5' and C_4', and C_3 generates the fused feature map C_3' by fusing the up-sampled C_6, C_5' and C_4' layers. In the same dimension, only the C_3', C_4', and C_5' layer

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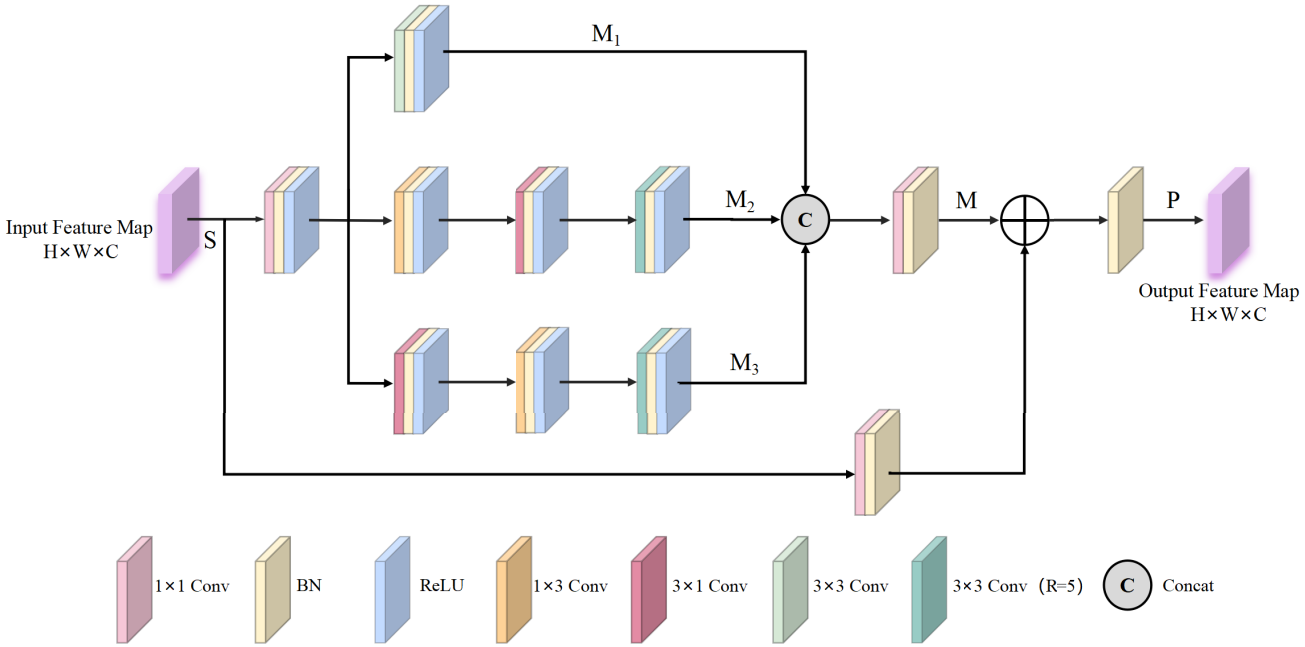


Fig. 3. Architecture of MRFFE.

feature maps are fused in the bottom–up path, and specifically, C_3' obtains the fused feature map C_3'' by fusing it with the residual-joined input C_3, C_4' obtains the fused feature map C_4'' by fusing with the convolved C_3'', C_4 of the residual link input, C_5' is obtained by fusing with the convolved C_3'', C_4'', and the residual connected input C_5 to obtain the fused feature map C_5''. This process can be computed as

$$C_i' = \mathrm{Conv}\left(\frac{C_i \times w_i + \delta_{\uparrow}^{2}\left(C_{i+1}'\right) \times w_{i+1} + \delta_{\uparrow}^{4}\left(C_{i+2}'\right) \times w_{i+2} + \ldots + \delta_{\uparrow}^{i+\max}\left(C_{i+\max}\right) \times w_{i+\max}}{w_i + w_{i+1} + w_{i+2} + w_{i+\max} + \varepsilon}\right) \tag{1}$$

$$C_i'' = \mathrm{Conv}\left(\frac{C_i \times w'_i + C_i' \times w_i'' + \delta_{\downarrow}^{2}\left(C_{i-1}''\right) \times w_{i-1}' + \ldots + \delta_{\downarrow}^{i-\min}\left(C_{i-\min}''\right) w_{i-\min}'}{w'_i + w_i'' + w_{i-1}' + \ldots + w_{i-\min}' + \varepsilon}\right) \tag{2}$$

where w_i, w_i', and w_i'' denotes different learnable weights for aggregating different levels of features, C_i denotes the multiscale feature map obtained in backbone, C_i' denotes the fused feature map generated during top–down fusion, and C_i'' denotes the fused feature map generated during bottom–up fusion. δ denotes a sampling operation, subscript $\uparrow$ for upsampling, subscript $\downarrow$ denotes convolutional downsampling, and superscript number indicates the sampling multiplier. i+max denotes the minimum-scale feature map (denoted by C_6 in this text), and i-min denotes the maximum-scale feature map (denoted by C_3'' in this text). ε =0.0001 is used to avoid numerical instability, and Conv denotes the convolution operation.

C. Multibranch Expanded Convolutional Fusion Mechanism

Small objects in optical images under aerial view are small in size and densely distributed. Traditional convolutional neural networks with a fixed receptive field find it difficult to capture sufficient information about the target objects, while the use of contextual information around the target can assist the network in obtaining more effective feature information. Expanding the receptive field by combining several large kernel convolutions of different sizes into the backbone network can capture more contextual information about the target. However, the introduction of large kernel convolution brings more background noise, which affects the small object detection accuracy, and also leads to the proliferation of the number of model parameters, which reduces the detection efficiency. Under the same dimensionality condition, dilated convolution can realize the technical requirement of introducing fewer parameters to obtain a larger receptive field in terms of capturing context information. Therefore, MRFFE is introduced in this article to obtain rich local context information. P_3 and P_4 are tasked with detecting small- and medium-sized objects, which are densely packed and have lower pixel. Inserting an MRFFE module at these stages enlarges the receptive field, captures richer contextual cues, and improves the separation of nearby objects. By contrast, P_5 and P_6 focus on large objects whose features are already well differentiated; attaching MRFFE here yields minimal accuracy gains yet adds unnecessary computation. Consequently, MRFFE is placed solely before P_3 and P_4 in the proposed architecture. Fig. 3 shows the detailed structure of MRFFE, which is

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mainly divided into four feature branches. The main branch is used to extract the feature map of contextual information, which consists of a 3×3 convolutional module and two dilated convolutional blocks with an expansion rate of 5. The residual branch is used to retain the small object key feature information and contains only residual blocks with a kernel size of 1×1.

The input to MRFFE is a feature map of size $H \times W \times C$. The input is divided into three main convolutional branches and one residual branch, and a 1×1 convolution operation is performed on all three main convolutional branches, keeping the feature map size constant. For convolutional branches, the first one uses a 3×3 convolutional module; the second one uses 1×3 and 3×1 convolutional modules successively, and finally feeds a 3×3 convolutional module with an expansion rate of 5; the third one is subjected to 3×1 and 1×3 convolutional modules, respectively, and the output results are fed into a 3×3 convolutional module with an expansion rate of 5. The output results of the three convolutional branches are then spliced along the channel dimensions and downscaled by using a 1×1 convolutional block, and the output feature maps are of size $H \times W \times C$. For the residual branch, a 1×1 convolution block is used, and the output feature map size is also $H \times W \times C$. The final output is generated by element-wise summation of the fused multibranch features and the residual features. This process can be formulated as

$$M_1 = \mathrm{Conv}_{3\times3}\left\{\mathrm{Conv}_{1\times1}(S)\right\} \tag{3}$$

$$M_2 = \mathrm{Conv}_{3\times3}^{r=5}\left\{\mathrm{Conv}_{3\times1}\left(\mathrm{Conv}_{1\times3}\left(\mathrm{Conv}_{1\times1}(S)\right)\right)\right\} \tag{4}$$

$$M_3 = \mathrm{Conv}_{3\times3}^{r=5}\left\{\mathrm{Conv}_{1\times3}\left(\mathrm{Conv}_{3\times1}\left(\mathrm{Conv}_{1\times1}(S)\right)\right)\right\} \tag{5}$$

$$M = \hat{Conv}_{1\times1}\{\mathrm{Cat}(M_1 + M_2 + M_3)\} \tag{6}$$

$$P = \mathrm{BN}\left\{M + \hat{Conv}_{1\times1}(S)\right\} \tag{7}$$

where $\mathrm{Conv}_{1\times1}$, $\mathrm{Conv}_{1\times3}$, $\mathrm{Conv}_{3\times1}$, and $\mathrm{Conv}_{3\times3}$ denote convolution operations with different convolution kernels. $\mathrm{Conv}_{3\times3}^{r=5}$ denotes a dilated convolution operation with a dilation rate of 5. $\hat{Conv}_{1\times1}$ denotes a convolution operation that contains only a 1×1 convolution module and a BN operation. Cat denotes a feature map splicing operation. S is the input feature map and P is the final output feature map.

IV. Experiment Results and Analysis

In this section, the experimental dataset, training setup, and evaluation metrics are described. Also, the experimental results are analyzed in detail.

A. Dataset

To evaluate the validity and superiority of the model architecture described in this article, simulation experiments are conducted using small-object remote sensing image datasets, including the VisDrone2019 dataset [53], the NWPUVHR-10 dataset [54], the SIMD dataset [55], and ablation experiments are conducted on the VisDrone2019 dataset. The details of each dataset are described below.

1) *VisDrone2019:* This dataset was collected by the AISKYEYE team at the Machine Learning and Data Mining Laboratory of Tianjin University, and the data covers UAV-captured images in different urban environments, with objects in the images covering ten categories: pedestrians, crowds, bicycles, cars, lorries, trucks, tricycles, awning tricycles, buses, and motorcycles. The total number of images is 10 209, which includes 6471 training images, 548 validation images, 1610 test images, and 1580 test challenge images. Among them, 1610 test images with annotations are used to test the model performance, and the object size distribution within the dataset is shown in Fig. 4.

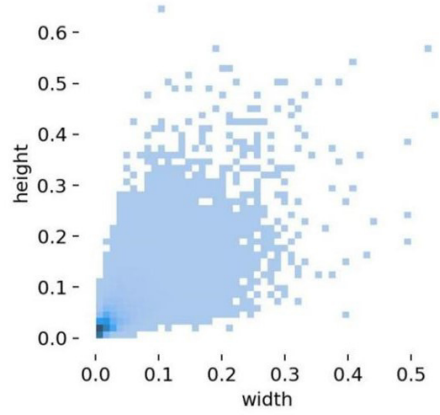


Fig. 4. VisDrone2019 dataset size distribution.

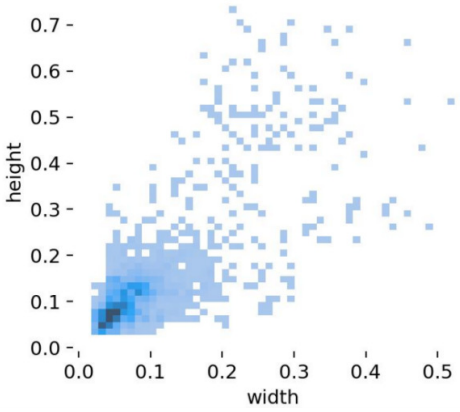


Fig. 5. NWPU VHR-10 dataset size distribution.

2) *NWPU VHR-10:* This dataset, released in 2014 by NWI, totals 800 high-resolution satellite images, including 650 object images and 150 background images. The images were taken from Google Earth and Vaihingen datasets, cropped, and manually labeled by experts. The object categories include aircraft, ships, storage tanks, baseball fields, tennis courts, basketball courts, surface runways, harbors, bridges, and vehicles. The images are partitioned into training (70%, 454 images), validation (5%, 32 images), and testing (25%, 164 images) subsets. The object size distribution within the dataset is shown in Fig. 5.
3) *SIMD:* This dataset was proposed by the National University of Science and Technology, Pakistan in 2020 and 15 categories are covered in the images namely: trucks, vans, cars, long vehicles, buses, airliners, propellers,

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TABLE I
RESULTS OF ABLATION EXPERIMENTS ON THE VISDRONE2019 DATASET

Methods	MDFFN	MRFFE	P(%)	R(%)	mAP50(%)	mAP50:95(%)	params
A1	×	×	49.9	37.3	38.1	22.7	11139470
A2	√	×	54.0	41.5	43.6	26.4	**8780010**
A3	×	√	53.6	41.6	43.3	26.2	10730440
A4	√	√	**54.3**	**42.9**	**44.5**	**27.2**	8835794

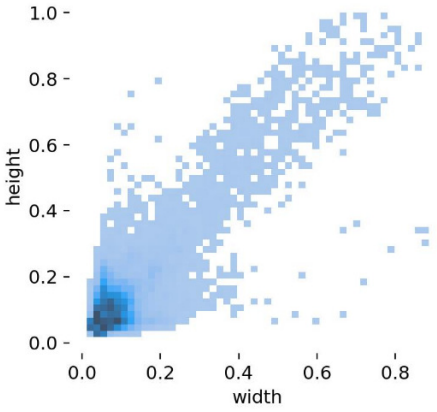


Fig. 6. SIMD dataset size distribution.

trainers, chartered, fighter, stair trucks, pushback truck, helicopter, boat, and others. The dataset consists of a total of 5000 images which are divided into 3000 for model training, 1000 for model validation, and 1000 for model testing. The object size distribution within the dataset is shown in Fig. 6.

B. Experimental Environment and Evaluation Metrics

The network was optimally learned using a stochastic gradient descent (SGD) optimizer with 200 epochs of training iterations on the VisDrone2019 and NWPU VHR-10 datasets with a training batch size of 8. The network was learned with an initial learning rate of 0.01, a momentum of 0.937, and weight decay of 0.0005. All experiments were run in a platform environment with NVIDIA 3090 GPUs, Python 3.8, PyTorch 1.12.0, and CUDA11.3.

Precision (precision, P), recall (recall, R), mean accuracy (mAP), parameter counts (params), and FLOPs (G) were selected as the standard evaluation metrics. Among them, P reflects the accuracy of model prediction, and R expresses the proportion of all positive categories correctly predicted by the model. mAP is divided into mAP50 and mAP50:95 according to different intersection and merger ratio (IoU) thresholds. mAP50 denotes the average precision value when the IoU threshold is 0.5, and mAP50:95 denotes the average precision value calculated with a step of 0.05 from 0.5 to 0.9. Detection accuracies were calculated for different IoU thresholds and averaged. Params and FLOPs reflect the complexity and computational requirements of the model.

C. Ablation Experiments

For verifying the effectiveness of each module in LMFF-MFFE, MDFFN and MRFFE were superimposed on the YOLOv8s [56] baseline algorithm, respectively, and experimental validation was carried out using the VisDrone2019 dataset. The experiments were conducted using single-factor variables to assess the impact of each improvement module on the detection algorithm. Table I shows the effect of adding or subtracting a module on the detection results, where √ means the module was used and × means it was not used.

As shown in Table I, model A2 with the addition of the MDFFN module showed a clear improvement in all the assessed metrics compared to the baseline algorithm A1. There was an increase of 5.5% and 3.7% on mAP50 and mAP50:95, respectively, and a 21.11% decrease in the number of parameters. The data demonstrated that MDFFN can improve detection performance by reducing the number of parameters and better balancing the dimensional relationship between detection speed and detection quality. Under the same experimental conditions, model A3 with the addition of the MRFFE module shows a clear improvement in all evaluation metrics compared to A1, especially in terms of the recall rate of 4.3%. This data indicate that the MRFFE module can effectively utilize contextual information to distinguish objects and reduce missed detection. Model A4 with both MDFFN and MDFFN modules added, that is, the model LMFF-MFFE proposed by us, shows an increase of 6.4% and 4.5% on mAP50 and mAP50:95, respectively, and a decrease of 20.4% in the number of parameters compared with A1. From the above analysis, it can be concluded that both the proposed MDFFN and MRFFE modules can improve the performance of LMFF-MFFE stably and there is no obvious conflict relationship between the modules.

The visualization results of A1, A2, A3, and A4 models are shown in Fig. 6, and the images show that both A1, A2, and A3 detection results have some degree of occurrence of missed detection (represented by the red solid rectangle box in Fig. 7) and misdetection (represented by the red dashed rectangle box in Fig. 7) and that the A4 model (LMFF-MFFE) can achieve robustness in the VisDrone 2019 dataset for objects of different scale objects with robust detection.

D. Comparative Experiments

To verify the superiority of LMFF-MFFE, the method of this article is compared with a batch of classical YOLO algorithms and current state-of-the-art methods for experiments, which are carried out under the same parameter conditions, with an input image size of 640×640, and trained on VisDrone2019 dataset and NWPU VHR-10 dataset.

On the VisDrone2019 dataset, the method of this article was compared with YOLOv10s/m [57], YOLOv11s/m [58], FFCA-YOLO [59], Mamba-YOLO-T/B [60],

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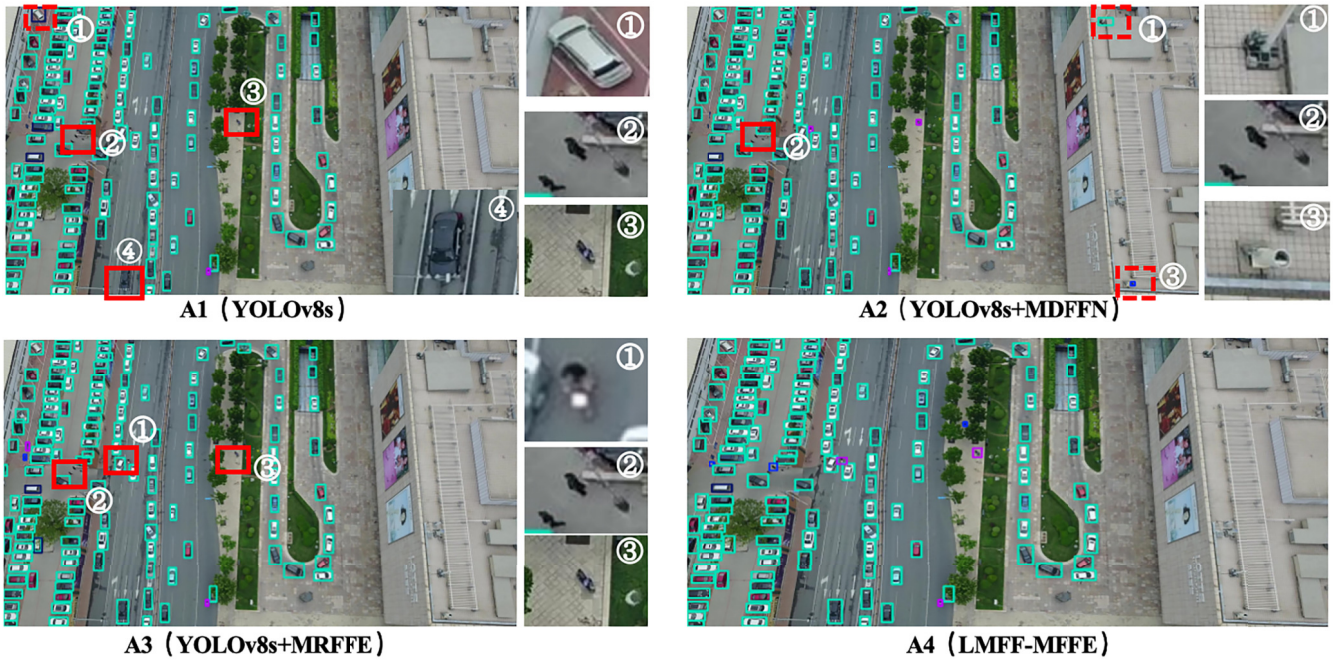


Fig. 7. Comparison of object detection visualization results for A1, A2, A3, and A4.

TABLE II

COMPARISON OF OBJECT DETECTION PERFORMANCE OF DIFFERENT METHODS ON THE VISDRONE2019 DATASET

Methods	P(%)	R(%)	mAP50(%)	mAP50:95(%)	params	GFLOPs
YOLOv8s	49.9	37.3	38.1	22.7	11.1	28.5
YOLOv10s	50.1	37.4	38.4	22.8	8.0	21.6
YOLOv10m	53.3	39.9	41.5	25.3	16.5	59.1
YOLOv11s	48.2	37.2	37.7	22.4	9.4	21.5
YOLOv11m	53.9	40.7	42.1	25.8	20.0	68.0
Mamba-YOLO-T	46.8	36.1	36.4	21.6	**6.0**	**13.6**
Mamba-YOLO-B	51.4	40.3	41.2	24.7	21.8	49.6
FFCA-YOLO	51.8	39.7	41.2	23.3	7.1	51.4
Mul-BiFPN+M-SimAM	46.9	38.0	36.4	20.1	-	-
DFFN	-	-	35.9	21.3	39.1	192
FPSOD	-	-	38.0	23.1	-	-
PS-YOLO	50.2	38.9	37.6	22.3	42.2	80.0
CFPT	-	-	38.4	22.6	56.3	297.6
LMFF-MFEE	**54.3**	**42.9**	**44.5**	**27.2**	8.8	41.7

TABLE III

COMPARISON OF OBJECT DETECTION PERFORMANCE OF DIFFERENT METHODS ON THE NWPU VHR-10 DATASET

Methods	P(%)	R(%)	mAP50(%)	mAP50:95(%)	params	GFLOPs
YOLOv8s	88.9	84.0	89.1	56.8	11.1	28.5
YOLOv10s	77.7	79.2	83.5	52.9	8.0	21.6
YOLOv10m	86.2	79.3	87.8	57.1	16.5	59.1
YOLOv11s	86.7	83.9	89.2	56.8	9.4	21.5
YOLOv11m	**92.6**	82.3	90.7	57.3	20.0	68.0
Mamba-YOLO-T	89.0	84.6	89.7	56.6	**6.0**	**13.6**
Mamba-YOLO-B	91.3	86.6	91.6	57.7	21.8	49.6
FFCA-YOLO	89.7	86.8	90.6	55.2	7.1	51.4
LMFF-MFEE	92.3	**87.1**	**92.6**	**58.1**	8.8	41.7

Mul-BiFPN+M-SimAM [61], PS-YOLO [62], CFPT [63], the FPSOD [64], and DFFN [65] are compared, and the comparison results are shown in Table II. Table II compares lightweight (YOLOv10s, YOLOv11s, and Mamba-YOLO-T) and medium-sized (YOLOv10m, YOLOv11m and Mamba-YOLO-B) variant. Lightweight models have fewer parameters and GFLOPs, but their detection accuracy is insufficient for small object detection. Compared with the lightweight model, medium models' accuracy has been improved, but at the cost of a marked rise in parameters and GFLOPs, making it difficult to achieve an effective balance between the model parameters and accuracy. By contrast, LMFF-MFFE achieves a markedly better balance: it uses even fewer parameters than YOLOv10s and Mamba-YOLO-T while delivering higher accuracy. Compared with FFCA-YOLO, DFFN, PS-YOLO, and CFPT, LMFF-MFFE improves by 3.3%, 8.6%, 6.9%, and 6.1% in terms of mAP50, respectively. The number of parameters as well as GFLOPs were significantly lower

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TABLE IV
COMPARISON OF OBJECT DETECTION PERFORMANCE OF DIFFERENT METHODS ON THE SIMD DATASET

Methods	P(%)	R(%)	mAP50(%)	mAP50:95(%)	params	GFLOPs
YOLOv8s	69.2	70.5	70.9	53.9	11.1	28.5
YOLOv10s	67.4	65.5	68.3	53.1	8.0	21.6
YOLOv10m	70.6	70.4	71.4	54.3	16.5	59.1
YOLOv11s	69.1	69.3	70.3	54.2	9.4	21.5
YOLOv11m	71.8	71.5	72.3	55.4	20.0	68.0
Mamba-YOLO-T	70.7	71.0	71.5	54.4	**6.0**	**13.6**
Mamba-YOLO-B	72.3	71.8	73.1	55.2	21.8	49.6
FFCA-YOLO	70.5	71.6	72.2	54.0	7.1	51.4
LMFF-MFEE	**73.9**	**72.3**	**74.3**	**55.8**	8.8	41.7

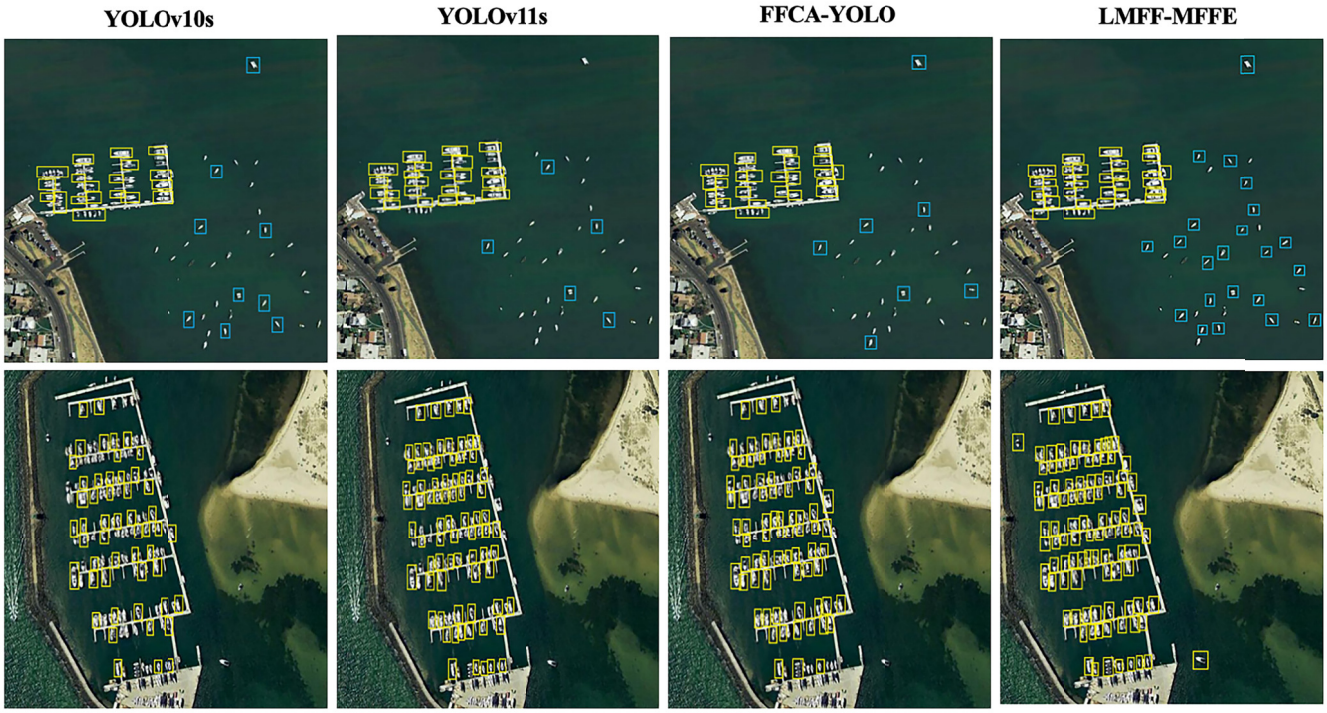


Fig. 8. Comparison of object detection visualization results for different detection methods.

than the above methods at mAP50:95 by 3.9%, 5.9%, 4.9%, and 4.6%, respectively. Compared with FPSOD as well as Mul-BiFPN+M-SimAM, LMFF-MFFE improves mAP50 by of 6.5% and 8.1% and mAP50:95 by 4.1% and 7.1%, respectively. The experimental results show that the method in this article has obvious advantages when targeting small object detection tasks in complex environments.

On the NWPU VHR-10 dataset, the method of this article is compared with YOLOv8s, YOLOv10s/m, YOLOv11s/m, FFCA-YOLO, and Mamba-YOLO-T/B. The results of the comparison are shown in Table III. Further, relative to YOLOv10s/m, YOLOv11s/m, and Mamba-YOLO-T/B, LMFF-MFFE achieves the best performance in small-object detection, demonstrating a better balance between accuracy and the cost. It improves 2.0% mAP50 and 2.9% mAP50:95 over the FFCA-YOLO, with 18.9% fewer GFLOPs. Compared to the baseline algorithm YOLOv8s, LMFF-MFFE increases by 3.5% mAP50 and 1.3% mAP50:95 while reducing parameters to 8.8M, demonstrating a superior accuracy-complexity trade-off.

Our method is compared with YOLOv8s, YOLOv10s/m, YOLOv11s/m, FFCA-YOLO, and Mamba-YOLO-T/B on the SIMD dataset. The results of the comparison are shown in Table IV. The accuracy improvement of YOLOv10s/m, YOLOv11s/m, and Mamba-YOLO-T/B is also accompanied by a rapid increase in the number of parameters and GFLOPs, making it difficult to achieve an effective balance between detection accuracy and computational complexity. In contrast, LMFF-MFFE performs best in small object detection, which outperforms the FFCA-YOLO method by 2.1% and 1.8% in mAP50 and mAP50:95, respectively. The dataset also shows that LMFF-MFFE can significantly improve small object detection accuracy with less computational complexity than the baseline algorithm, achieving an effective balance between accuracy and model complexity.

By observing the experimental results on multiple datasets, it is found that the algorithm LMFF-MFFE in this article is able to obtain high detection accuracy with less computational complexity. The result is driven by two critical parts in the model, that is, MDFFN and MRFFE. MDFFN enhances the ability of small-object feature representation by introducing multiple cross-layer dense connections and jumping connections to deeply fuse the information of multilayer feature maps, avoiding the information loss. However, the additional con-

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nections and feature interactions in LMFF-MFFE inevitably increase computational demands compare to the baseline YOLOv8s. To balance fusion quality and efficiency, we prune insignificant feature propagation paths while preserving detection accuracy. This jointly enhances feature map fusion and suppresses computational growth. MRFFE adopts multibranch dilation convolution, and due to the unique nature of dilation convolution, it can obtain rich contextual information around the small object without increasing the number of additional parameters, which improves the detection accuracy of the small object.

Since our model is based on the YOLOv8s algorithm to improve the proposed LMFF-MFFE, considering representativeness and attention of the algorithms in the field of small object detection, this article visualizes the detection results of YOLOv10s, YOLOv11s, FFCA-YOLO, and LMFF-MFFE as shown in Fig. 8. The visualization results show that YOLOv10s, YOLOv11s, and FFCA-YOLO have different degrees of leakage when performing the small object detection task, and the detection results are unsatisfactory. LMFF-MFFE, on the other hand, exhibits good robustness, and can fully utilize the context information to achieve dense small object detection with a low leakage rate. The experimental results show that LMFF-MFFE has good performance in performing the small object detection task.

V. Discussion

To evaluate how well LMFF-MFFE balances accuracy and model size, this study compared it with leading detectors on three datasets: VisDrone2019, NWPU VHR-10, and SIMD. Lightweight versions of YOLOv10, YOLOv11, and Mamba-YOLO lose substantial accuracy, while their full models recover performance only at the cost of a large increase in parameters and FLOPs, indicating deficient in balancing the accuracy and complexity. Likewise, recent methods such as FFCA-YOLO, PS-YOLO, and DFFN improve small-object detection accuracy but still exhibit an unfavorable accuracy-to-complexity trade-off. In contrast, LMFF-MFFE achieves the best performance on all three datasets with lower computational complexity. Compared to the baseline, LMFF-MFFE can significantly improve the detection accuracy with a small increase in computational complexity. Future work will extend this model to multimodal and lightweight based on UAV scenarios by fusing infrared, radar, and other multisource data to achieve more accurate and robust small object detection without sacrificing efficiency.

VI. Conclusion

To address the challenges of low accuracy and high computational complexity in small object detection, this article proposes a LMFF-MFFE algorithm. First, a lightweight MDFFN is designed, which integrates multiscale features by combining shallow fine-grained details with deep semantic information, enriching feature representations while reducing parameters. Second, to enhance the multiscale applicability of the fused features, a MRFFE is used to assign different weights to the features of different receptive fields, to obtain the local context information related to the small objects, and to improve the differentiation of the objects. Finally, a small object detection head is added to improve the model's ability to perceive small objects. Experimental results on the VisDrone2019 and NWPU VHR-10 datasets demonstrate LMFF-MFFE's superior accuracy–complexity trade-off, outperforming the current mainstream methods.

While this study focuses on unimodal lightweight small object detection, future research will explore multimodal lightweight frameworks by designing efficient cross-modal fusion networks. These networks aim to integrate multimodal data while minimizing computational overhead, further advancing deployability in resource-constrained UAV platforms.

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